ThaliSense

Eat the food you love.
Know what it does to you.

An on-device AI nutrition and metabolic-health coach built for Indian meals.

SHAKTI Hackathon · Obesity Control & Medical Fitness · AI-Agentic

Live: multihacks-unstop26.vercel.app

github.com/andringodson/Hackathon-MultiHacksUnstop26

PROBLEM

India’s diabetes and obesity crisis is cooked in everyday kitchens, and our tools don’t speak its language.

101 M
Indians living with diabetes

136 M
more with prediabetes, mostly undiagnosed

BMI 23
is where overweight starts for Asian Indians, not the global 25

- Calorie apps are built for Western diets. They don't know pesarattu, bisi bele bath or kadhi, or what “do roti aur dahi” means.
- Risk is mis-measured. South Asians get diabetes at lower weight, so global BMI tells at-risk people they're fine.
- “Eat less rice” doesn't stick, dietitians are unaffordable, and people won't upload health data to a server.

Source: ICMR-INDIAB, Lancet Diabetes & Endocrinology 2023; Misra et al., JAPI 2009

PROPOSED SOLUTION

A coach that understands Indian food, measures risk the Indian way and plans meals you'll actually eat.

1 · Log

Type it your way ("rendu dosai", "2 roti, dal") or snap a photo. On-device AI recognises 108 Indian dishes.

2 · Understand

Indian Diabetes Risk Score, Asian BMI cut-offs, waist-to-height ratio and personal calorie and protein targets.

3 · Coach

An agent finds what matters most and swaps foods you already eat, showing the weekly impact of each swap.

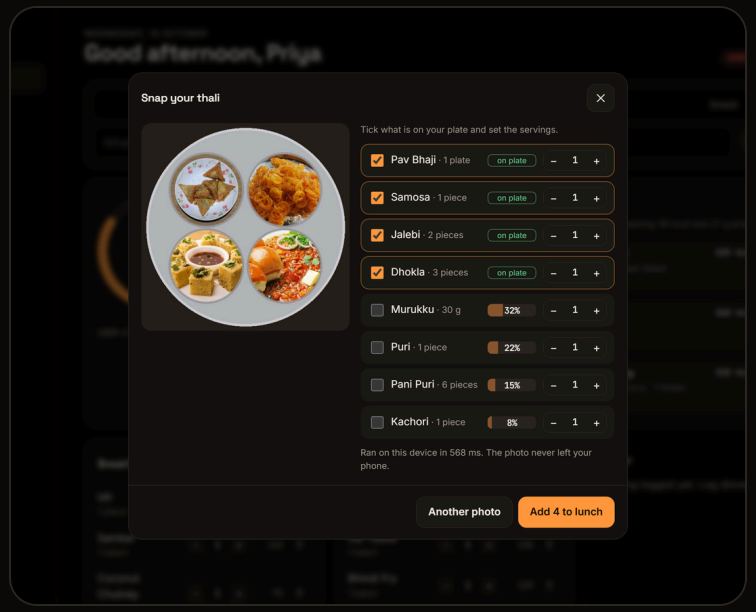
4 · Plan

A self-checked 7-day regional meal plan for your diet and ₹/day budget. Log any meal with one tap.

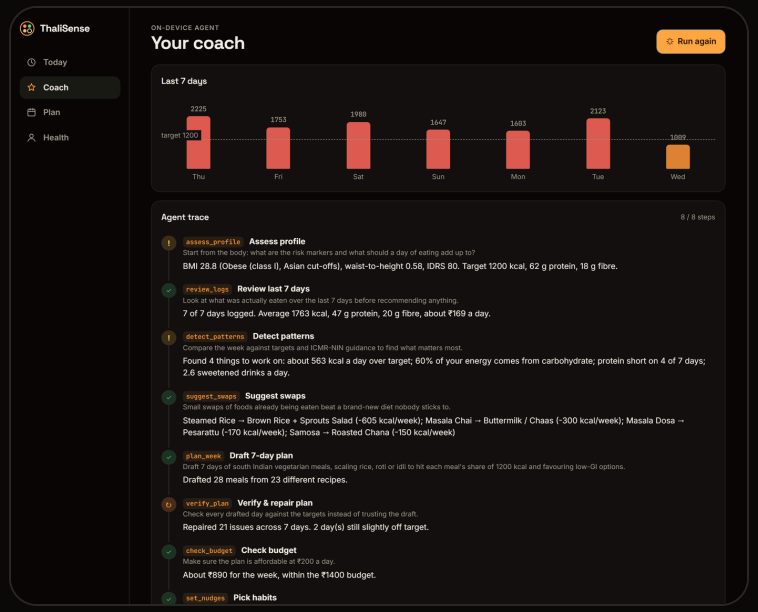
Free. No sign-up. Nothing leaves your phone: the AI runs inside the browser.

WORKING MVP · LIVE TODAY

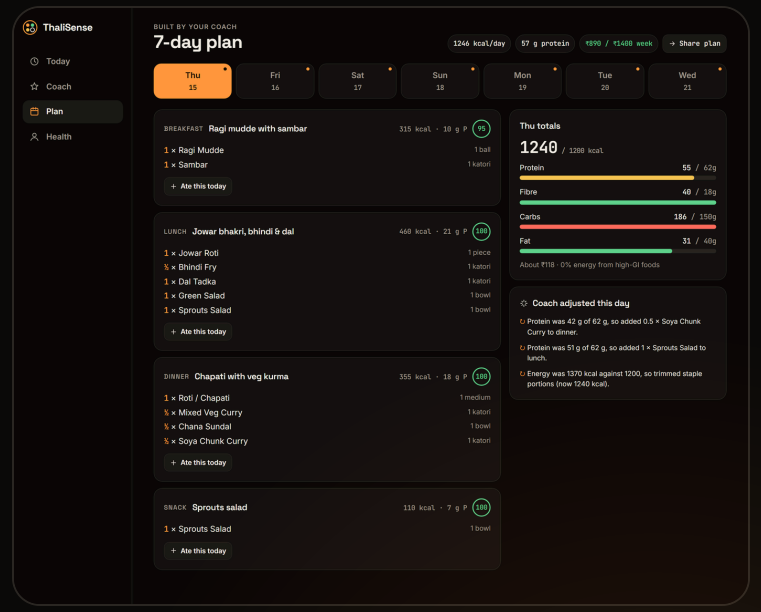
From a photo of your thali to a week of better meals.



Snap a whole thali: the AI scans the plate and ticks every dish it finds, on the device.



Coach: an 8-step agent trace you can follow, plus insights and swaps.



Plan: 7 regional days within budget, with notes on what the coach adjusted.

Try the demo week at multihacks-unstop26.vercel.app · Thali photo: composite of Wikimedia Commons images, CC BY-SA 4.0 (credits in the repo)

ThaliSense

TECHNOLOGY · AI USED

Real AI, running entirely on the user's phone.

input
Typed meal / photo
→
parser · MobileCLIP-S2
Dish recognition
→
localStorage
Meal log
→
coach agent
8 tool steps
→
output
Insights · swaps · plan

Vision: fine-tuned MobileCLIP on the phone

Transformers.js on ONNX Runtime Web, WebGPU or 4-thread WASM, in a Web Worker. One vector per dish from descriptive prompts, few-shot adapted on 2,354 labelled photos. The 72 MB encoder downloads once; a second pass scans five plate regions to find every dish on a thali.

Agent: plan, verify, repair

A tool-based planner: `assess → review → detect → swap → plan → verify → budget → habits`. It scores 60 real meal templates on protein, fibre, glycaemic load, cost and region, scales the staples, then checks every day against the targets and fixes the misses.

React 19 · TypeScript · Vite · Transformers.js · ONNX Runtime Web · Vitest · Playwright + axe · GitHub Actions · Vercel · PWA · no backend, no paid APIs

MEASURED, NOT GUESSED

We benchmarked four vision models and fine-tuned the best.

MODEL, ON 1,694 TEST PHOTOS	SIZE	TOP-1	TOP-3
CLIP ViT-B/32 (our v1)	89 MB	71.5%	86.8%
CLIP ViT-B/16	87 MB	73.6%	87.9%
MobileCLIP-S2	72 MB	81.5%	92.9%
MobileCLIP-S2, few-shot adapted	**72 MB**	**87.8%**	**95.0%**

25% → 55%

of dishes found on 200 four-dish test thalis, by adding a scan of five plate regions to the whole-photo guess.

Test photos from three public Indian-food datasets, never used for training or tuning. Full model card: github.com/andringodson/Hackathon-MultiHacksUnstop26/blob/main/docs/vision.md

IMPLEMENTATION APPROACH

Built from scratch for SHAKTI, grounded in Indian clinical science.

How we built it

- Food data: 108 dishes from IFCT 2017 (ICMR-NIN) with serving sizes and cost
- Risk: MDRF IDRS, Asian BMI consensus, Mifflin-St Jeor, ICMR-NIN 2024
- Vision: 4 models benchmarked on 1,694 photos; MobileCLIP-S2 few-shot adapted
- Agent: template scoring with a verify-and-repair loop; 17 unit tests
- Shipped: 12 end-to-end tests incl. accessibility, CI on every push, auto-deploy

Status & what is new

Everything shown was built during the hackathon window. No pre-built project code was reused; open-source libraries (React, Transformers.js), the public MobileCLIP model and public datasets are credited.

Typed logging, 108 dishes	Live
On-device photo and whole-thali recognition	Live
Risk scores and targets	Live
Coach agent, 7-day plan, plate scores, progress	Live
Plate segmentation, voice in Indian languages	Next

EXPECTED IMPACT

Small swaps, caught early, at national scale.

~1,200

kcal a week saved by the swaps the coach finds in the demo week, about 0.7 kg of fat a month, with more protein and fibre.

60+

IDRS score triggers a clear “get tested” prompt, catching undiagnosed diabetes earlier.

0

bytes of health data uploaded. Privacy builds trust, especially for women and older users.

- Fights obesity with the food people already eat, not a foreign diet they'll quit in a week.
- Inclusive by design: vegetarian-aware, budget-aware (₹200 a day by default), regional, and runs on low-end phones.
- Scales at near-zero cost: static hosting, on-device AI, works offline.

FUTURE SCOPE

Where ThaliSense goes next.

- Plate segmentation: outline each katori, then recognise it, for crowded thalis.
- Voice logging in Hindi, Tamil, Bengali and Marathi.
- More dishes adapted: labelled photos for all 108 dishes, then the next 200.
- ASHA-worker mode: offline community IDRS screening and referrals.
- Wearable and glucometer import for steps and glucose response.
- Optional LLM chat, grounded in the agent's trace, with the user's own key.

ThaliSense

Try it now.

Open the link on any phone and press **Try with a demo week**.

Live app

multihacks-unstop26.vercel.app

Source code

github.com/andringodson/Hackathon-MultiHacksUnstop26